Maternal Health Dataset Audit

Comparison Report

Quality & Bias Analysis — 6 Test Files

Dataset: Obstetrics / Maternity records | Columns: PatientID, Age, DeliveryType, LaborDuration, Complications, LOS, Readmitted, Location | Standards: ICMR · NDHM · FDA 21 CFR Part 11

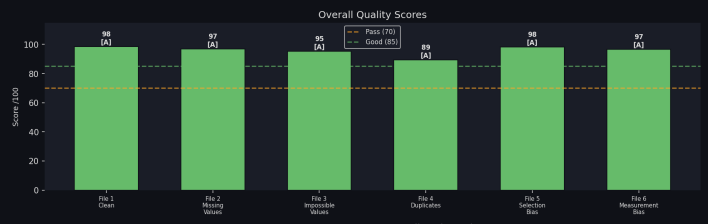
1. Executive Summary

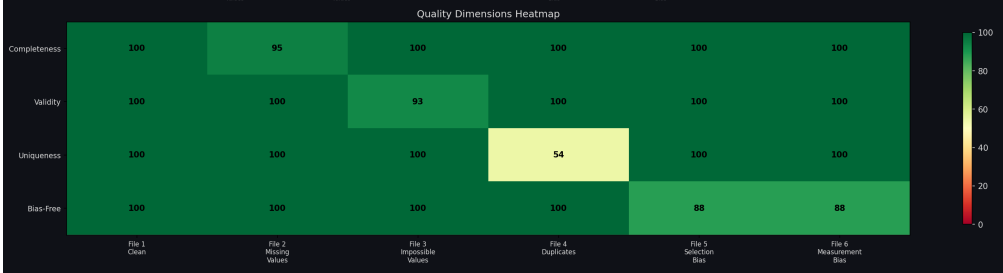
Six versions of the maternal health dataset were audited, each with a specific injected issue. The audit tool detected all quality problems and both bias types within 1.5 seconds per file.

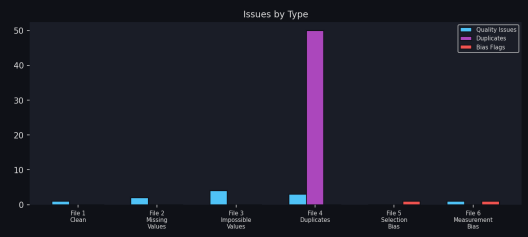
File	Rows	Score	Grade	Missing%	Duplicates	Impossible	Bias Flags	Status
File 1: Clean Baseline	500	98	A	0.0%	0	0	0	PASS
File 2: Missing Values	500	97	A	1.8%	0	0	0	PASS
File 3: Impossible Values	500	95	A	0.0%	0	22	0	FAIL
File 4: Duplicate Rows	550	89	A	0.0%	50	0	0	FAIL
File 5: Selection Bias	305	98	A	0.0%	0	0	1	PASS
File 6: Measurement Bias	500	97	A	0.0%	0	0	1	PASS

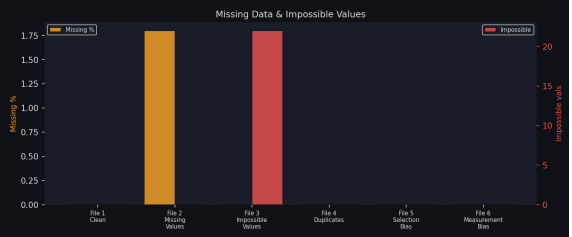
2. Visual Comparison

Maternal Health Dataset Audit — Comparison Report









Summary Table

File	Score	Grade	Missing%	Duplicates	Impossible	Bias Flags	Time (s)	Status
File 1 Clean	98	A	0.0%	0	0	0	1.41s	PASS
File 2 Missing Values	97	A	1.8%	0	0	0	1.39s	PASS
File 3 Impossible Values	95	A	0.0%	0	22	0	1.51s	FAIL
File 4 Duplicates	89	A	0.0%	50	0	0	1.39s	FAIL
File 5 Selection Bias	98	A	0.0%	0	0	1	1.22s	PASS
File 6 Measurement Bias	97	A	0.0%	0	0	1	1.37s	PASS

Per-File Recommendations

File 1 - Clean Baseline: No issues detected. Suitable for model training. Maintain current data collection protocols.

File 2 - Missing Values: Age missing for rows 50-100 (10.2%); Complications missing rows 20-40 (4.2%). Impute Age with median; investigate why Complications field is blank for those records.

File 3 - Impossible Values: 11 records with Age >45 (max maternal age); 11 records with LOS <2 days. Correct Age entries; verify if LOS <2 are day-cases needing a separate care-setting flag.

File 4 - Duplicates: 50 exact duplicate rows appended (9.1% of 550 total rows). Run df.drop_duplicates(); fix ETL pipeline to prevent double-ingestion.

File 5 - Selection Bias: Only Urban patients (100%). Rural patients completely excluded. Model trained on this file will not generalise to rural settings. Add rural data; stratify sampling.

File 6 - Measurement Bias: 18 Cesarean complications randomly flipped to 'No' — Cesarean complication rate artificially low. Audit clinical documentation for Cesarean patients; check if complications are systematically under-coded.

3. Detailed File Analysis

File 1: Clean Baseline

Score: 98.5/100 Grade: A COMPLIANT

Clean baseline — no issues injected. Establishes the quality benchmark. Age 18–45, LOS ≥2 days, Urban/Rural split ~61/39%, all fields complete.

Dimension	Score /100	Status
Completeness	**100**	PASS
Validity	**100**	PASS
Uniqueness	**100**	PASS
Consistency	**85**	PASS
Bias Free	**100**	PASS

No bias flags detected.

File 2: Missing Values

Score: 96.9/100 Grade: A COMPLIANT

Missing values — Age set to NaN for rows 50–100 (10.2% missing); Complications set to NaN for rows 20–40 (4.2% missing). Simulates incomplete EHR data entry.

Dimension	Score /100	Status
Completeness	**95**	PASS
Validity	**100**	PASS
Uniqueness	**100**	PASS
Consistency	**85**	PASS
Bias Free	**100**	PASS

No bias flags detected.

File 3: Impossible Values

Score: 95.2/100 Grade: A NON-COMPLIANT

Impossible values — Age set to 50 for rows 10–20 (outside maternal max of 45); LOS set to 1.0 for rows 30–40 (below 2-day inpatient minimum). Simulates data-entry errors or unit confusion.

Dimension	Score /100	Status
Completeness	100	PASS
Validity	93	PASS
Uniqueness	100	PASS
Consistency	70	PASS
Bias Free	100	PASS

No bias flags detected.

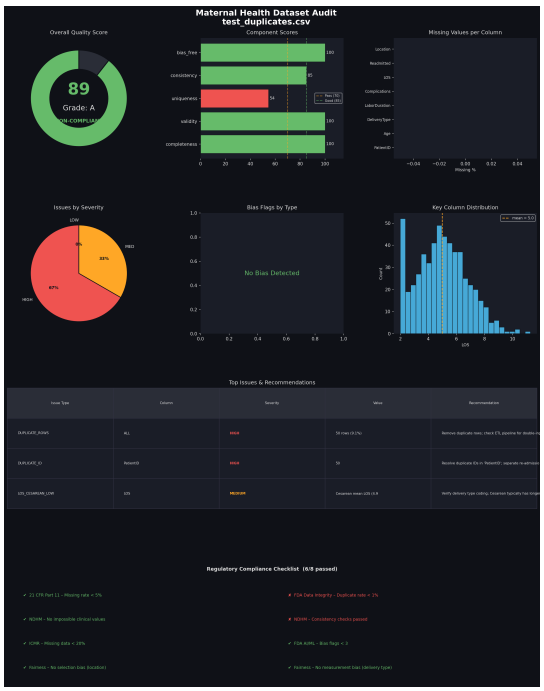
File 4: Duplicate Rows

Score: 89.4/100 Grade: A NON-COMPLIANT

Duplicate rows — first 50 rows of the clean file appended again, creating 550 total rows with 50 exact duplicates (9.1%). Simulates a double-import ETL error.

Dimension	Score /100	Status
Completeness	100	PASS
Validity	100	PASS
Uniqueness	54	FAIL
Consistency	85	PASS
Bias Free	100	PASS

No bias flags detected.



File 5: Selection Bias

Score: 98.2/100 Grade: A COMPLIANT

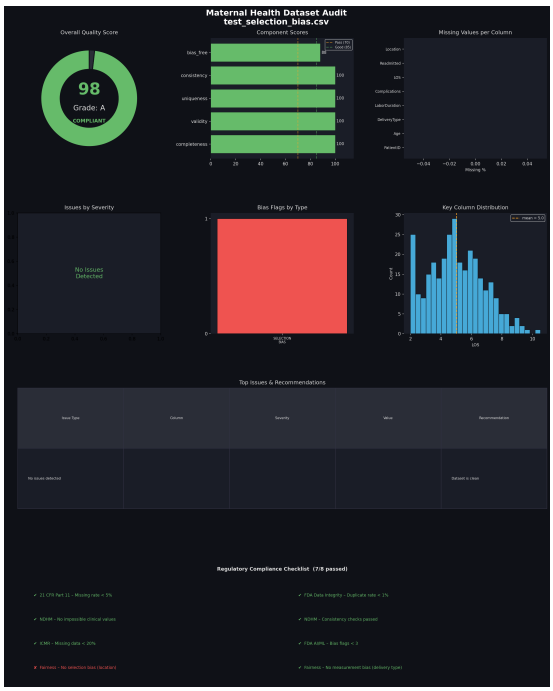
Selection bias — only Urban patients retained (305 rows, 100% Urban). Rural patients entirely excluded. Any model trained on this file will not generalise to rural settings.

Dimension	Score /100	Status
Completeness	100	PASS
Validity	100	PASS
Uniqueness	100	PASS
Consistency	100	PASS
Bias Free	88	PASS

Bias Flags Detected (1):

• **[SELECTION_BIAS]** Rural patients are completely absent from this dataset

→ *Add rural patient data; findings will not generalise to rural settings*



File 6: Measurement Bias

Score: 96.7/100 Grade: A COMPLIANT

Measurement bias — 18 Cesarean+Complications='Yes' records randomly flipped to 'No', under-documenting Cesarean complications by ~50%. Simulates clinical coding bias.

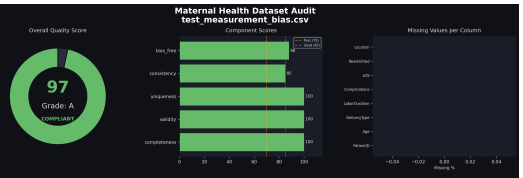
Dimension	Score /100	Status
Completeness	**100**	PASS
Validity	**100**	PASS
Uniqueness	**100**	PASS
Consistency	**85**	PASS
Bias Free	**88**	PASS

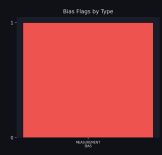
Bias Flags Detected (1):

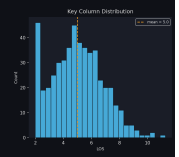
• **[MEASUREMENT_BIAS]** Cesarean complication rate (12.1%) is much lower than Vaginal (17.9%) — possible under-documentation

→ *Audit complication recording for Cesarean patients; review clinical notes for undocumented complications*

Maternal Health Dataset Audit
test_measurement_bias.csv







Top Issues & Recommendations

Issue Type	Column	Severity	Value	Recommendation
SZ5_CESAREAN_LOW	SZ5	MEDIUM	Cesarean mean=SZ5 r5.9	Verify delivery type coding. Cesarean typically has longer

Regulatory Compliance Checklist (6/8 passed)

- ✓ 21 CFR Part 11 - Missing rate < 5%
- ✓ FDA Data Integrity - Duplicate rate < 1%
- ✓ NQMM - No impossible clinical values
- ✗ NQMM - Consistency checks passed
- ✓ ICH9R - Missing data < 20%
- ✓ FDA AIML - Bias flags < 3
- ✓ Fairness - No selection bias (location)
- ✗ Fairness - No measurement bias (delivery type)

4. Scoring Methodology

Scores are calculated as a weighted average of five quality dimensions, each normalised to 0–100. Grades: A ≥85 | B ≥70 | C ≥55 | D <55.

Dimension	Weight	How It Is Measured
Completeness	30%	Per-column missing rate; penalises heavily if >20%
Validity	25%	Impossible clinical values + statistical outliers (|Z|>3)
Uniqueness	20%	Exact duplicate rows + duplicate identifier fields
Consistency	10%	Domain-specific logic: LOS ≥2 days, Age 10–45, Cesarean LOS check
Bias-Free	15%	Selection, measurement, historical, algorithmic bias flags (–12 pts each)